

ROLLING HILLS AGCM, AB, Canada

WMO: 711050

Lat: 50.2650N

Lon: 111.7011W

Elev: 645

StdP: 93.81

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.8	-27.2	-34.1	0.2	-30.7	-30.4	0.2	-27.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	32.0	17.0	30.0	16.6	28.2	16.1	18.8	27.9	17.9	27.1	17.0	26.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	12.0	22.3	14.5	11.1	20.9	13.5	10.4	19.7	56.0	27.9	52.8	27.1	50.1	26.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-36.5	35.6	3.0	1.7	-38.7	36.8	-40.4	37.8	-42.1	38.8	-44.3	40.0
			-36.6	21.1	3.0	1.0	-38.8	21.9	-40.5	22.5	-42.2	23.0	-44.4	23.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-8.1	-9.0	-1.6	5.1	11.1	15.4	18.6	17.9	12.9	5.6	-2.2	-9.3
	DBStd	11.89	8.87	8.77	8.15	4.93	4.31	3.22	2.75	3.19	4.64	5.20	7.67	8.82
	HDD10.0	2795	561	532	362	158	38	2	0	1	23	153	367	598
	HDD18.3	5041	819	764	619	398	226	98	31	48	169	396	616	856
	CDD10.0	887	0	0	1	10	73	164	266	244	111	17	2	0
	CDD18.3	91	0	0	0	0	2	10	38	33	8	0	0	0
	CDH23.3	2152	0	0	0	12	126	261	771	714	258	10	0	0
	CDH26.7	733	0	0	0	2	27	71	271	271	91	1	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	328	14	11	15	25	43	70	41	36	30	16	15	13
	PrecMax	486	31	21	37	72	96	186	112	75	85	44	40	29
	PrecMin	154	5	3	1	3	6	14	6	2	5	3	3	2
	PrecStd	69	7	5	9	14	24	38	29	19	20	10	10	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	11.2	18.1	25.1	29.5	32.1	34.0	34.3	32.6	24.9	18.6	9.9
		MCWB	5.3	5.1	8.6	11.6	14.3	17.4	18.2	17.7	15.8	13.3	9.7	4.7
	2%	DB	8.1	7.8	15.5	21.1	26.9	28.6	31.7	31.9	29.6	20.7	13.9	6.9
		MCWB	3.7	3.2	7.3	10.0	13.6	16.0	17.8	16.7	15.0	11.0	7.3	3.2
	5%	DB	6.0	5.2	12.6	18.0	24.2	26.2	29.6	29.8	26.3	17.8	11.0	4.9
		MCWB	2.7	1.7	5.7	8.1	12.4	15.2	17.3	16.6	14.0	9.6	5.7	1.9
	10%	DB	4.1	2.9	9.3	15.2	21.4	24.2	27.8	27.6	23.3	15.0	8.3	3.1
		MCWB	1.5	0.5	4.1	6.6	11.2	14.3	16.9	16.0	12.8	8.1	3.8	0.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	6.0	9.2	11.7	15.4	18.9	20.7	19.5	17.1	13.7	9.8	5.5
		MCDB	9.9	11.1	16.7	22.6	27.2	27.9	29.3	28.7	28.6	22.4	17.6	9.3
	2%	WB	4.0	3.4	7.8	10.2	14.0	17.5	19.3	18.2	15.6	11.7	7.7	3.4
		MCDB	7.7	7.3	14.5	20.2	25.0	26.5	27.7	28.3	26.8	19.3	13.2	6.6
	5%	WB	2.7	2.0	6.0	8.6	13.0	16.1	18.3	17.3	14.5	10.0	5.9	2.0
		MCDB	5.7	5.0	12.0	16.4	22.4	24.3	26.9	27.2	25.0	17.1	10.6	4.9
	10%	WB	1.6	0.5	4.2	7.2	11.9	14.9	17.5	16.5	13.3	8.6	4.1	0.6
		MCDB	4.1	2.8	9.1	14.4	19.9	22.6	26.1	25.8	22.4	14.4	7.8	3.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	12.0	13.0	14.7	15.1	14.2	16.2	16.5	16.1	14.4	12.1	10.8
		MCDBR	11.2	14.1	17.2	21.2	20.3	18.5	19.9	21.3	22.3	18.5	15.7	11.7
		MCWBR	8.0	10.1	10.1	10.8	8.9	8.0	8.5	8.5	9.6	9.7	9.7	8.7
	5% WB	MCDBR	11.3	13.6	16.7	19.2	17.9	17.1	17.9	18.8	20.8	17.9	15.4	11.9
		MCWBR	8.2	10.0	10.1	10.1	8.0	8.0	8.3	8.0	9.0	9.7	9.9	9.0
(40) Clear-Sky Solar Irradiance	taub		0.277	0.282	0.310	0.364	0.384	0.372	0.371	0.397	0.347	0.305	0.276	0.264
	taud		2.382	2.441	2.417	2.262	2.259	2.349	2.370	2.267	2.393	2.487	2.464	2.383
	Ebn at Noon		771	871	901	877	872	885	879	834	846	830	774	732
	Edh at Noon		62	76	94	123	129	119	115	121	96	72	57	54
(44) All-Sky Solar Radiation	RadAvg		1.21	2.20	3.55	4.92	5.84	6.33	6.88	5.54	3.96	2.44	1.35	0.94
	RadStd		0.12	0.19	0.25	0.34	0.41	0.38	0.34	0.35	0.34	0.26	0.11	0.08

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page